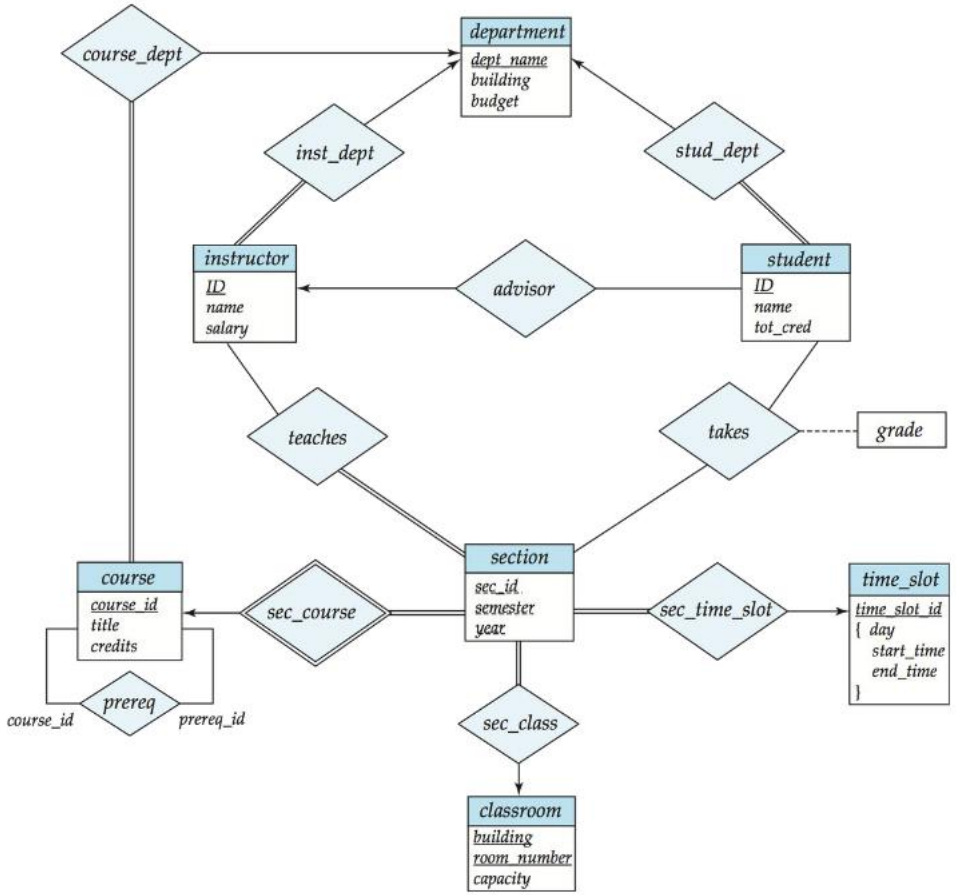
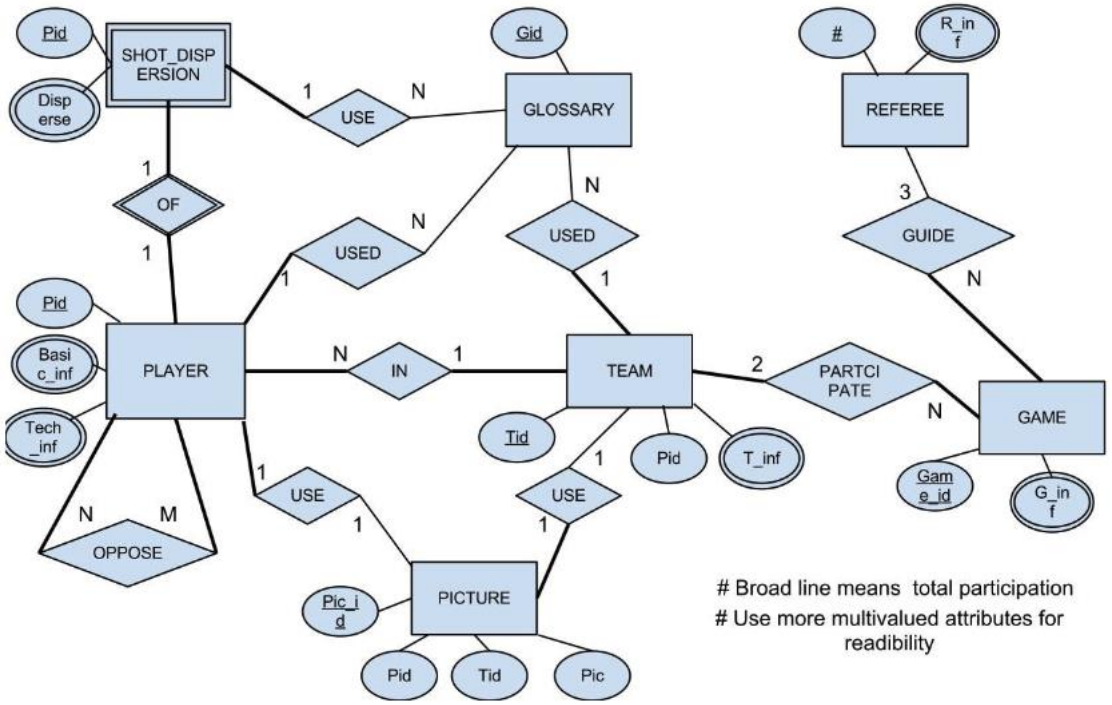
Entity Relationship Modeling

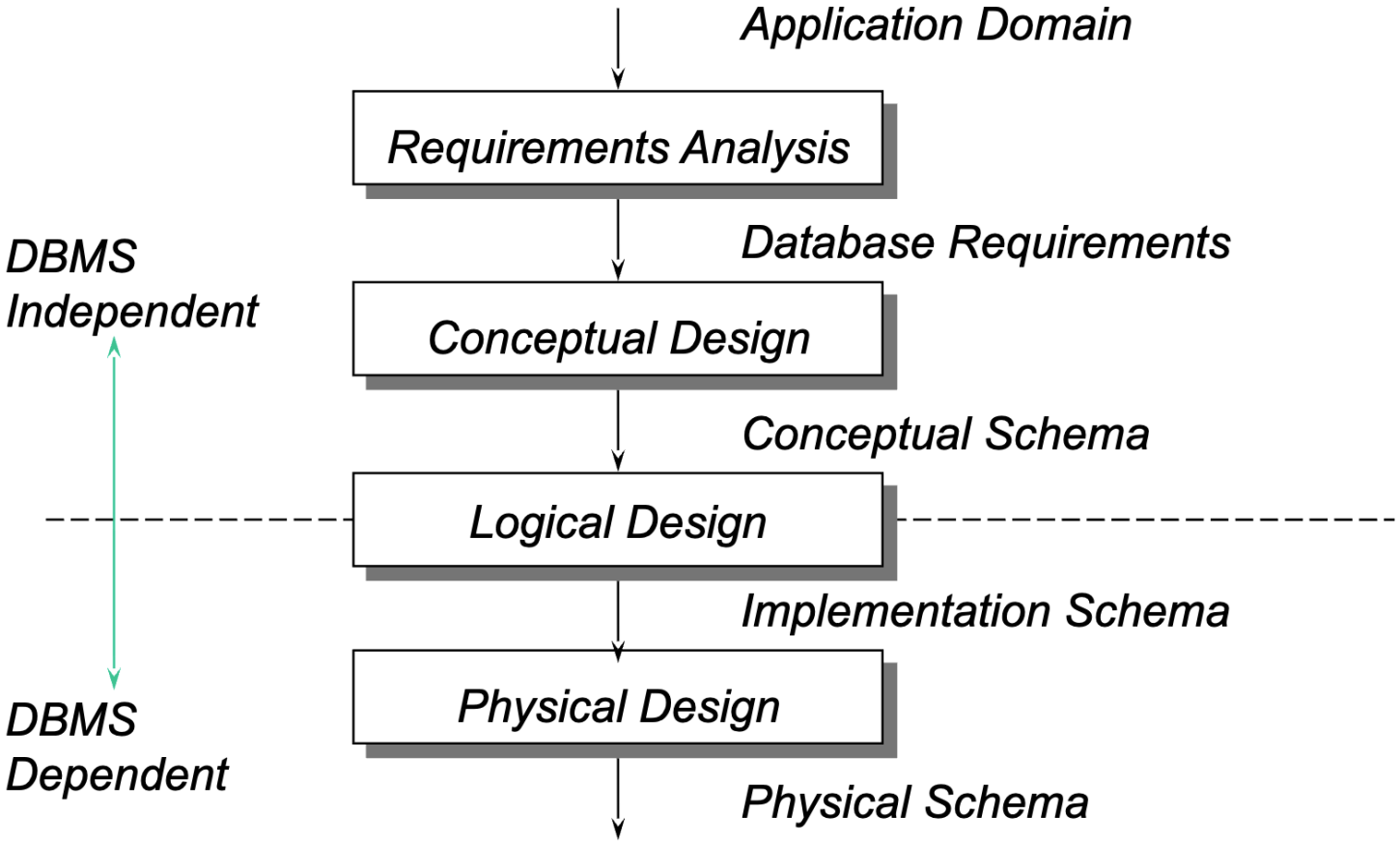
Instructor: Kazi Amit Hasan





Many diagrams taken from: Database Illuminated by Ricardo, Urban, Davis

Phases of Database Design



Conceptual Design

- Description of the database in a high-level model
  - close to user's view of world
  - semantic model – conveys meaning as well as structure
  - DBMS independent & remains correct even if the DBMS is changed
  - create a permanent description of requirements

Do you jump straight to coding for group projects?

Discussion

Requirement and feasibility analysis

Very very high level structure >> for DB, it's ER model

Many other steps

Coding

What will happen if you directly jump to coding?

Entity-Relationship Model

- Most popular conceptual model
- Developed by Peter Chen in 1976
- Exclusively used for conceptual design
- A database is modeled as a collection of entities, attributes and relationships among entities.
- Represented using ER diagrams

Let’s break down the title!

Entity Relationship Model

Any object that has physical existence

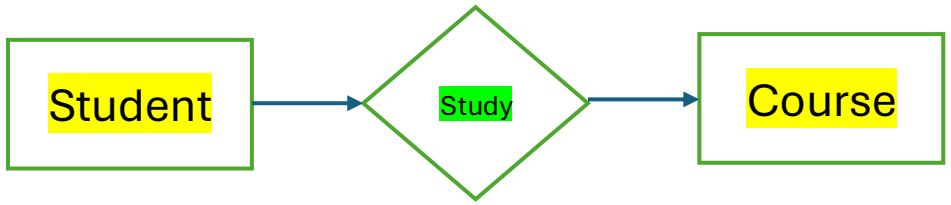
Student
- ID
- Name
- Age
- GPA

← Attributes

Association among entities

Course
- ID
- Name
- Duration
- Instructor

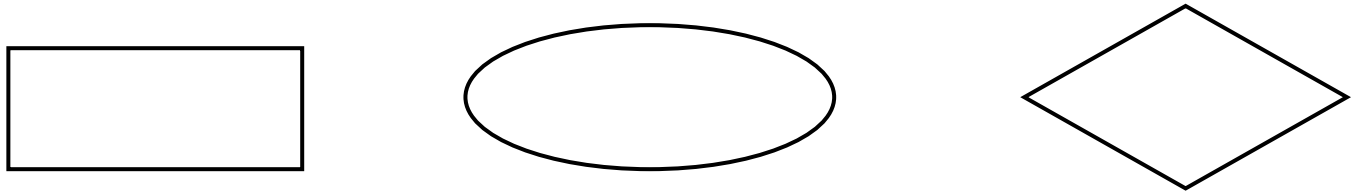
← Attributes



Entity-Relationship Model

- Entity
  - Attributes and it’s types
- Relationships
  - Types of relationships

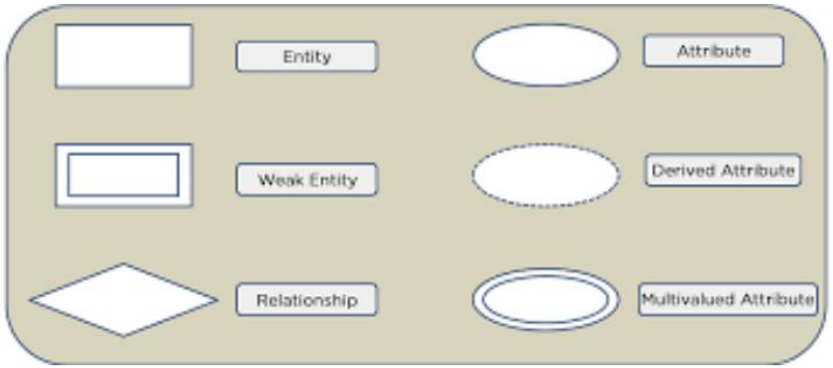
Representation



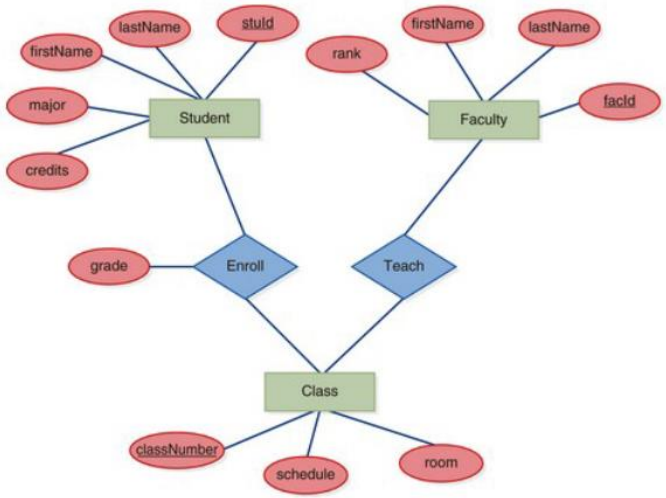
Entity Attributes Relationships

Model vs Schema

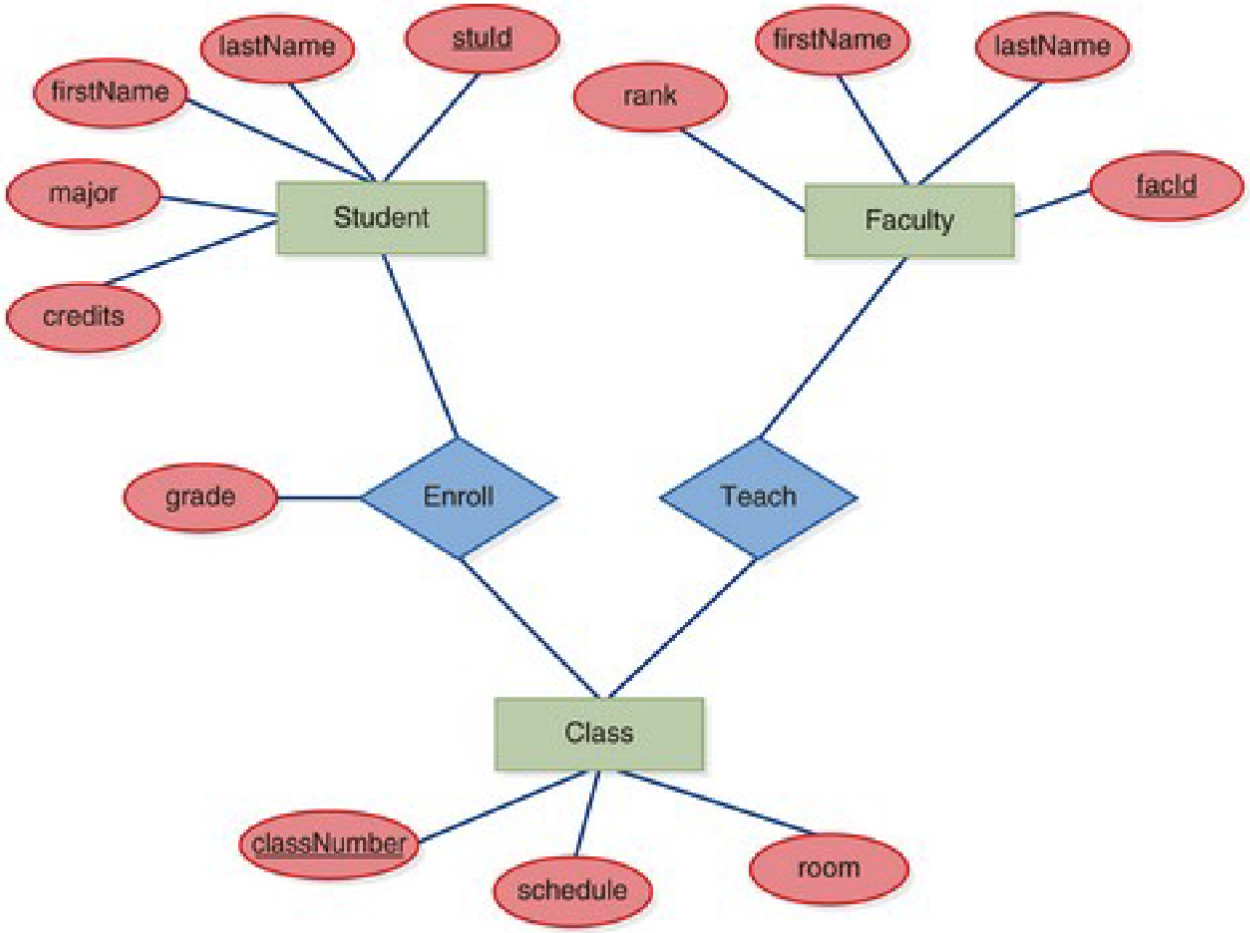
ER Model



ER Schema



Is this making sense (a bit) now?

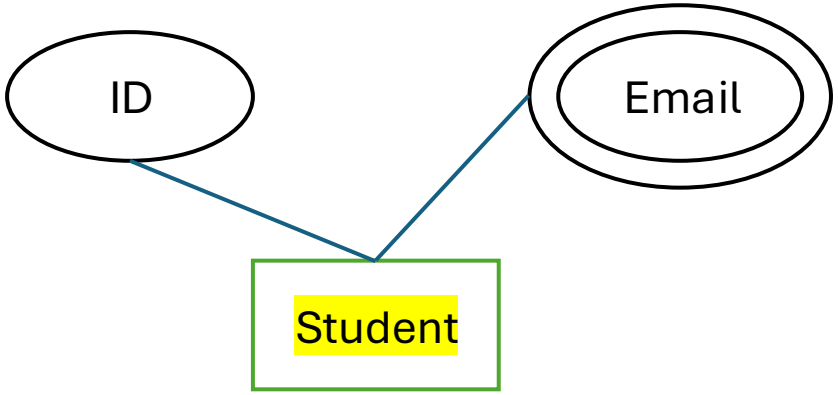


Entities

- An entity is an object that exists and is distinguishable from other objects.
  - Ex: People, companies, courses
- An entity instance is a particular instance of an entity
  - Ex: a particular student

Attributes

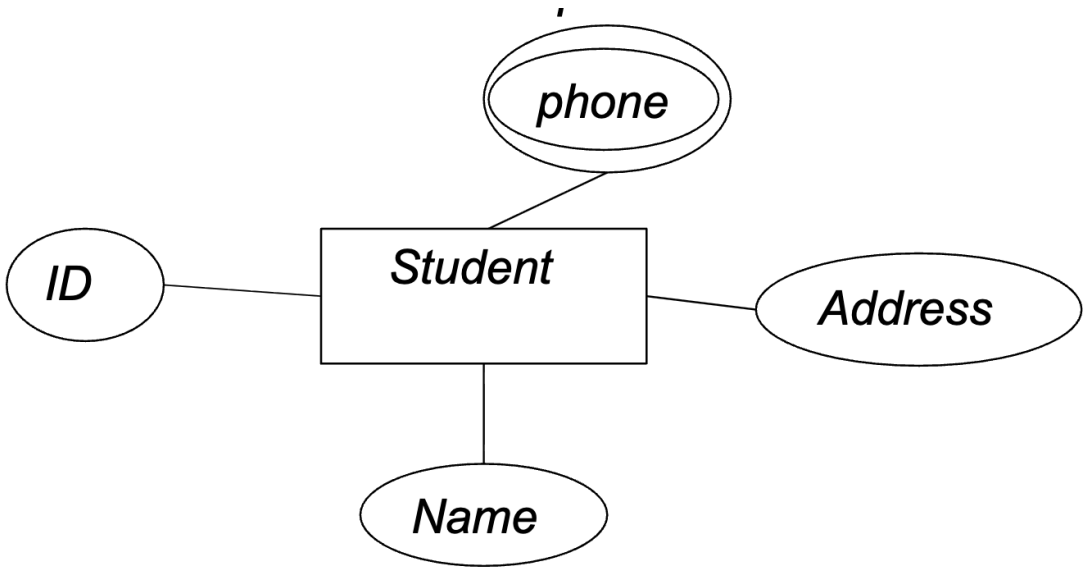
- Entities have attributes
  - Ex: People have IDs, names, addresses, DOBs
- Some attributes may be multi-valued (vs single valued)
  - Single valued >> only one value
    - ID
  - Multi-valued >> can be more than one values.
    - email



What others can be single and multivalued attributes?

Composite Attributes

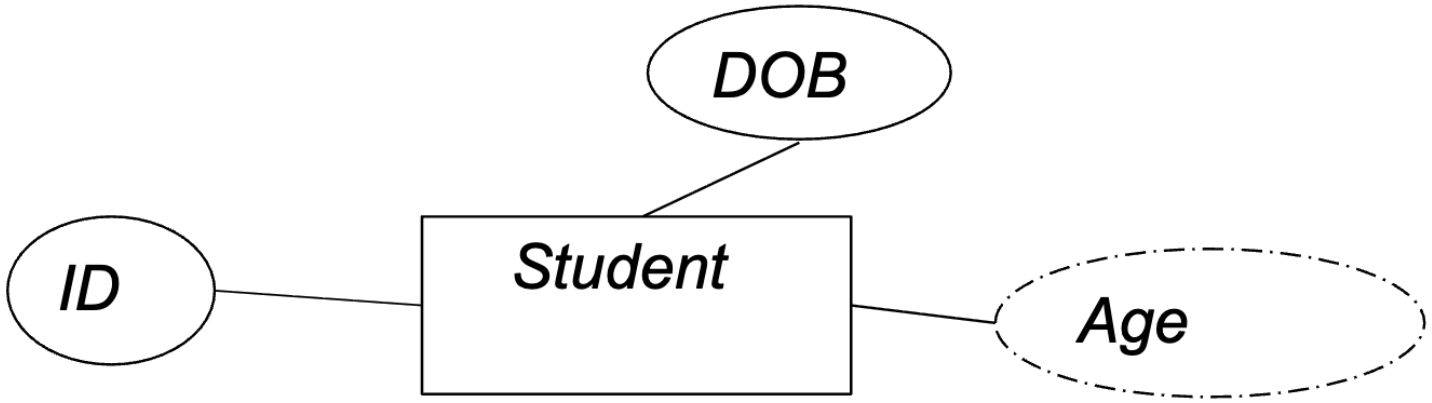
- Decompose an attribute into smaller parts



Which attributes can be composite here?

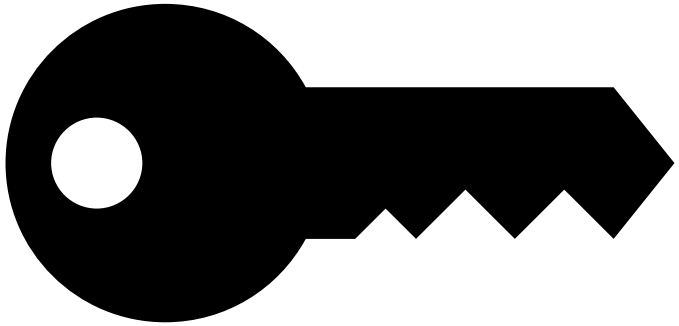
Derived Attribute

- Derived attributes get their values via calculation
- They are not stored in the DB, but instead ‘derived’ when they are requested.



What is derived attribute here? And how?

Keys

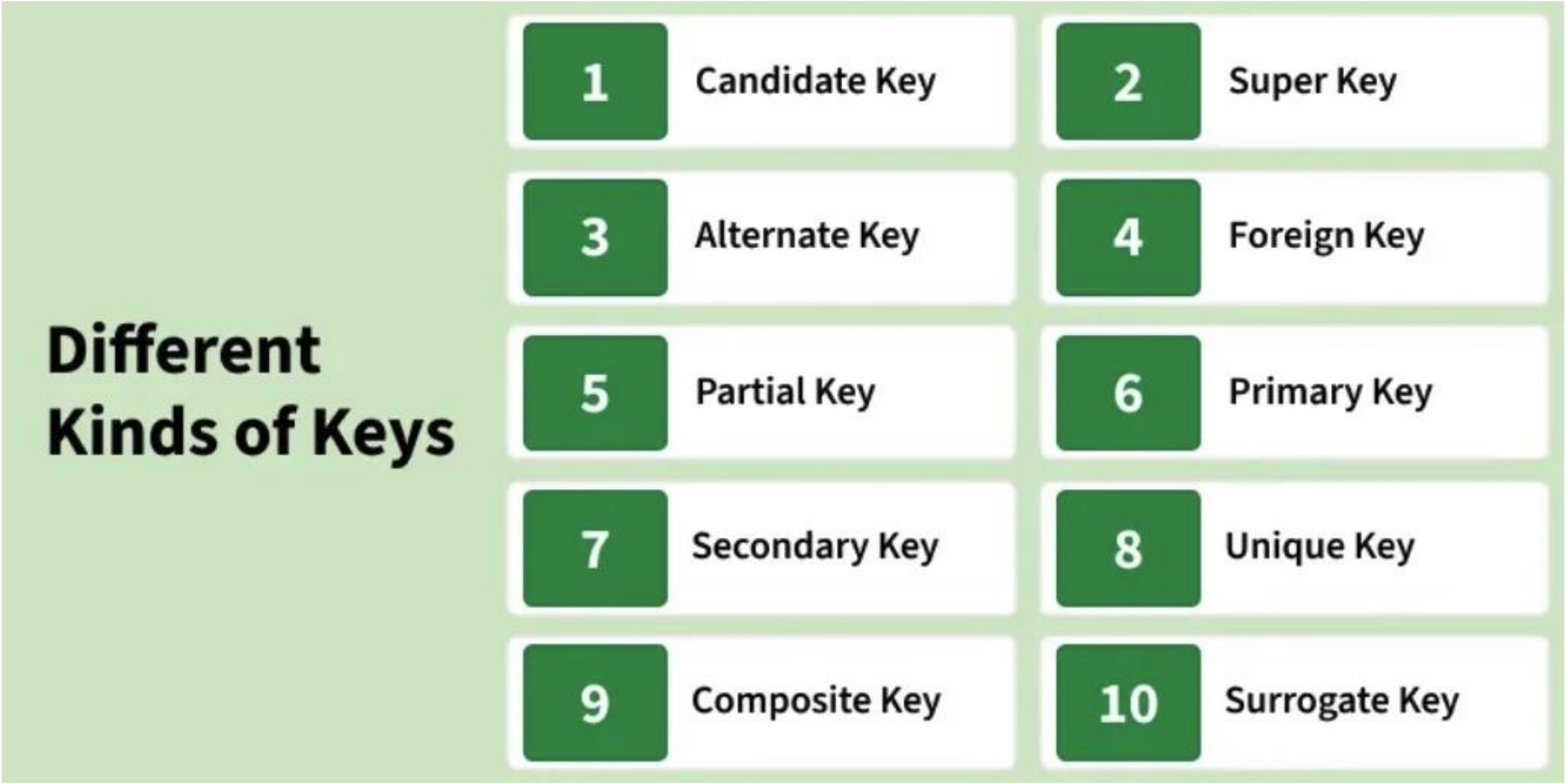


What is a key?

Keys

- Entity sets (the set of all instances for an Entity) contain unique instances – we have to be able to distinguish between them!
- Keys:
  - An attribute or a set of attributes
  - a data item (or set of items) that allow us to tell entity instances apart.
- Use of key:
  - Uniquely identify a row/tuple in a table
  - Ensures data integrity and enable efficient data retrieval
  - Without keys, it would be extremely difficult to manage large datasets and queries would become inefficient and prone to errors.

Different types of keys



Superkey

- A super key is a group of single or multiple keys that uniquely identifies rows in a table.
- A super key can contain extra attributes that aren't necessary for uniqueness.
- For example, if the "STUD_NO" column can uniquely identify a student, adding "SNAME" to it will still form a valid super key, though it's unnecessary.

STUD_NO	SNAME	ADDRESS	PHONE
1	Tom	New York	123456789
2	John	Los Angeles	223365796
3	Alice	Chicago	175468965

A super key could be a combination of STUD_NO and PHONE, as this combination uniquely identifies a student.

Candidate key

- A candidate key is a minimal super key, meaning it can uniquely identify a record but contains no extra attributes.
- It is a super key with no repeated data is called a candidate key.
- The minimal set of attributes that can uniquely identify a record.
- A candidate key must contain unique values, ensuring that no two rows have the same value in the candidate key's columns.

STUD_NO	SNAME	ADDRESS	PHONE
1	Tom	New York	123456789
2	John	Los Angeles	223365796
3	Alice	Chicago	175468965

STUD_NO can be a candidate key, as it uniquely identifies each record.

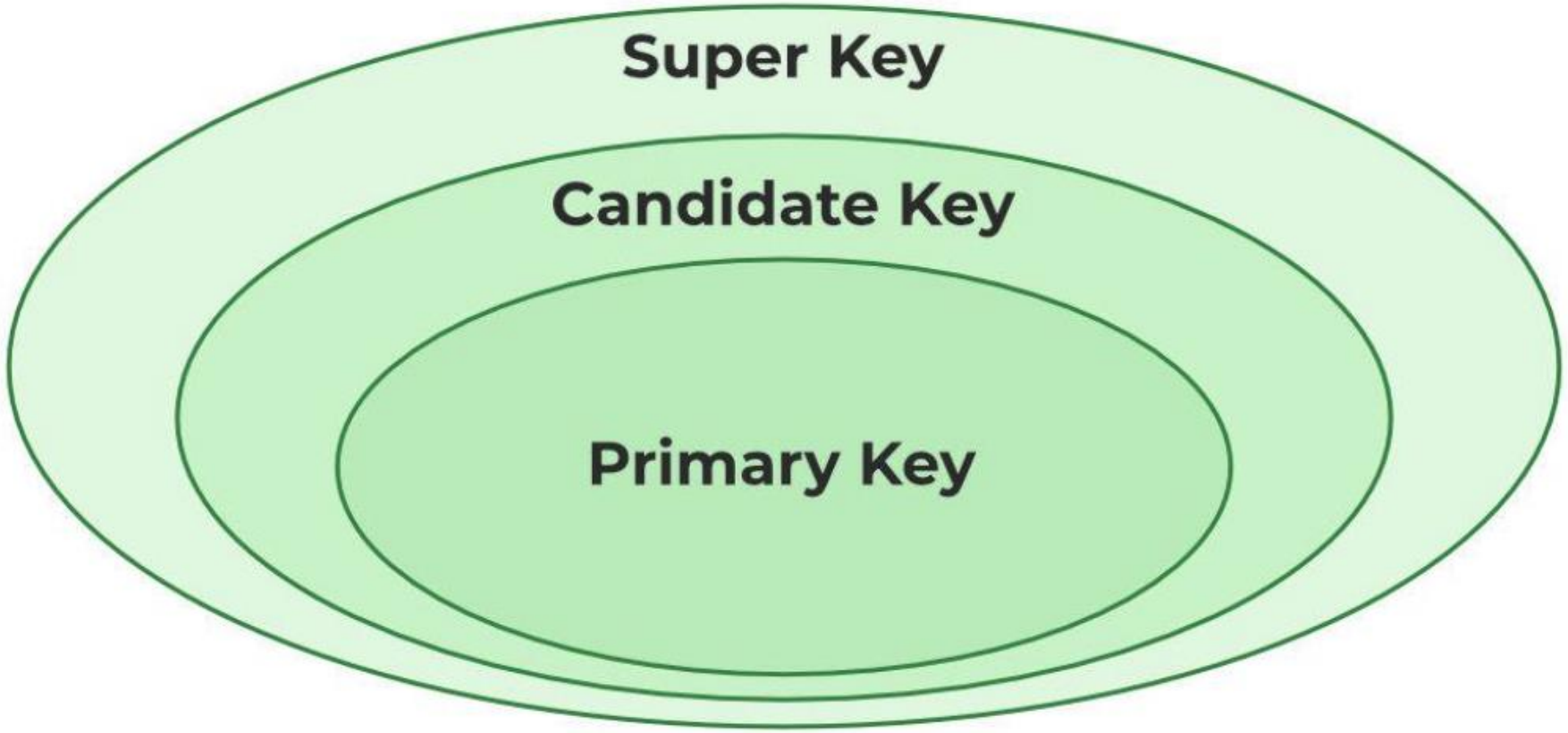
Primary key

- A primary key is a unique key, meaning it can uniquely identify each record (tuple) in a table.
- It must have unique values and cannot contain any duplicate values.
- A primary key cannot be NULL, as it needs to provide a valid, unique identifier for every record.
- Databases typically store rows ordered in memory according to primary key for fast access of records using primary key.
- The primary key for each entity is underlined in the ER mode

STUD_NO	SNAME	ADDRESS	PHONE
1	Tom	New York	123456789
2	John	Los Angeles	223365796
3	Alice	Chicago	175468965

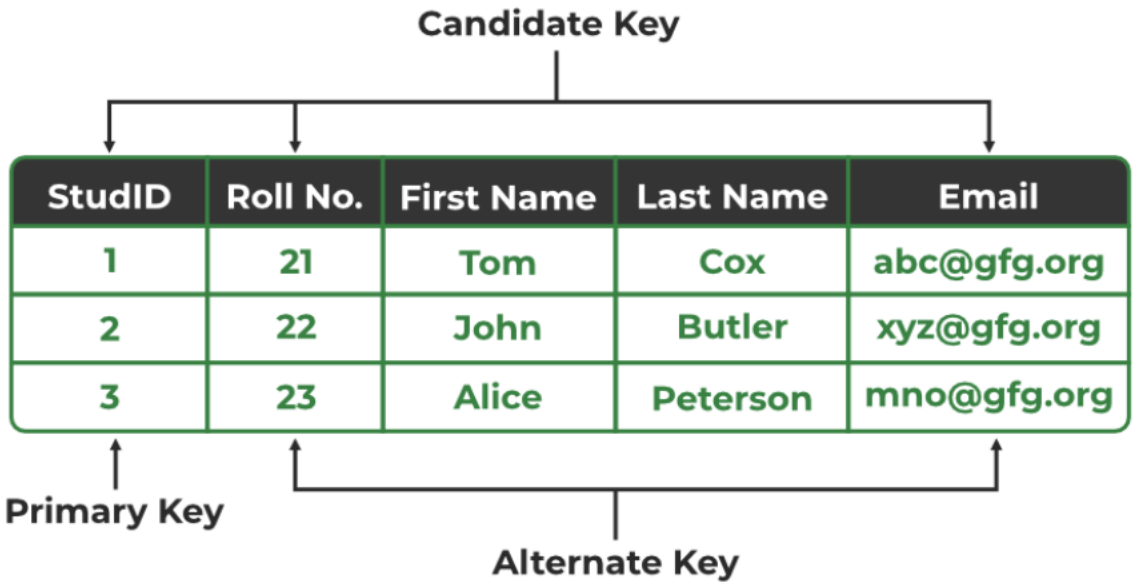
<u>STUD_NO</u> is a primary key

Relation among Primary Key, Candidate Key, and Super Key



Alternate key

- An alternate key is any candidate key in a table that is not chosen as the primary key.
- An alternate key can consist of one or more columns (fields) that can uniquely identify a record, but it is not the primary key



Composite Keys

- a candidate key that consists of more than one attribute
- Ex:
  - *Customer id* needs to be entered every time the order is placed
  - *Customer id* can repeat in the table and can not serve as a primary key
  - *Product id* and *Quantity* cannot be chosen as primary key because multiple customers can buy same product and the same quantity.

Customer Id	Product Id	Quantity
121	9034	50
333	9446	67
121	9869	30
201	9446	67
223	9023	80

[Customer id + Product id] can be used to identify the records uniquely.

Composite Keys

- a candidate key that consists of more than one attribute
- Ex:
  - *Customer id* needs to be entered every time the order is placed
  - *Customer id* can repeat in the table and can not serve as a primary key
  - *Product id* and *Quantity* cannot be chosen as primary key because multiple customers can buy same product and the same quantity.

Composite Key

Customer id	Product id	Quantity
121	9034	50
333	9446	67
121	9869	30
201	9446	67
223	9023	80

[Customer id + Product id] can be used to identify the records uniquely.

Try yourself!

Name	Nationality	City	Age
Salah	Egyptian	Liverpool	32
Mane	Senegalese	Riyadh	30
Gakpo	Dutch	Liverpool	27
Salah	Egyptian	Liverpool	32

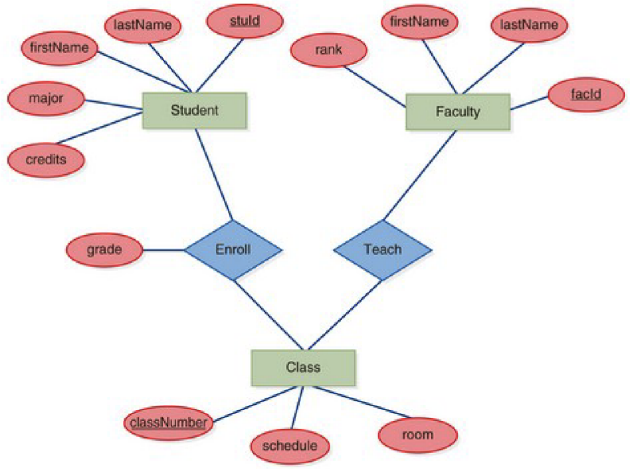
What’s the problem with this table?

PlayerID	RegID	PassportNo	Email	Phone	Name	Nationality	City	Age
1	R101	P8901	salah@mail.com	555-1111	Salah	Egyptian	Liverpool	32
2	R102	P6723	mane@mail.com	555-2222	Mane	Senegalese	Riyadh	30
3	R103	P4456	gakpo@mail.com	555-3333	Gakpo	Dutch	Liverpool	27
4	R104	P9987	salah2@mail.com	555-4444	Salah	Egyptian	Liverpool	32

- Identify the Super Keys
- Identify the Candidate Keys
- Identify the Primary Key
- Identify the Alternate Keys
- Identify any Composite Keys

Relationships

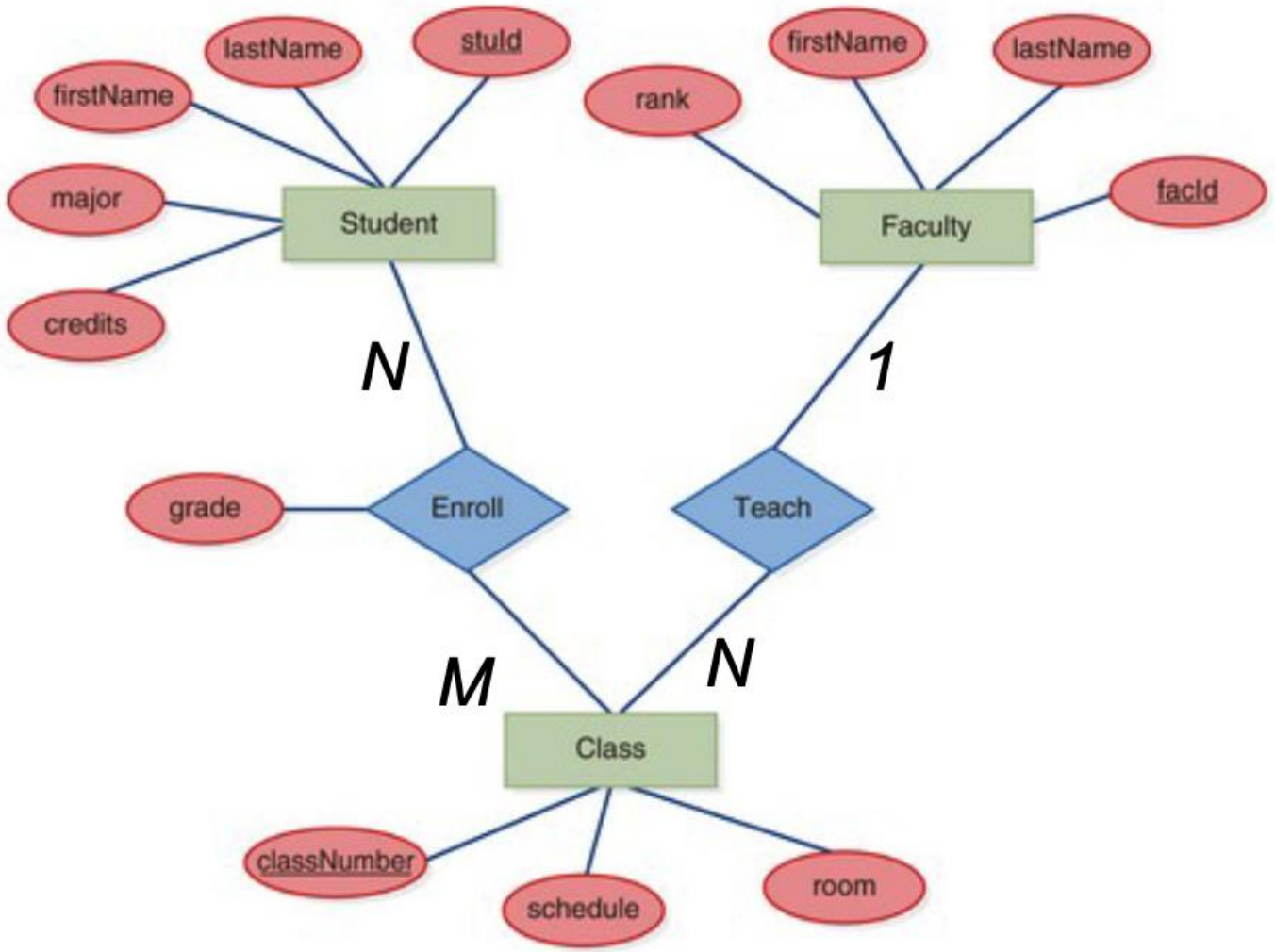
- A relationship is an association between entities
- relationships may have attributes associated with them



Cardinality Constraints

- Express the number of entities to which another entity can be associated via a relationship set
- The cardinality will be one of the following types:
  - One to one (1:1)
  - One to many (1:N)
  - Many to many (N:M)

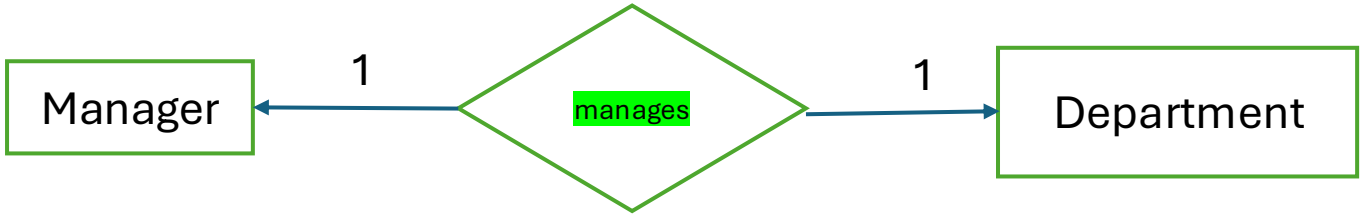
Cardinality Constraints



Can you explain this?

One to one relationship (1:1)

- each entity instance of X is associated with only one entity instance of Y



Explain this!

This indicates that an instructor may advise at most one student, and a student may have at most one advisor

One to Many Relationship (1:N)

- each entity instance of X may be associated with many entity instances of Y but each entity instance of Y is associated with at most one instance of X

This indicates that an instructor may advise many students, but a student may have at most one advisor.

Many to Many Relationship

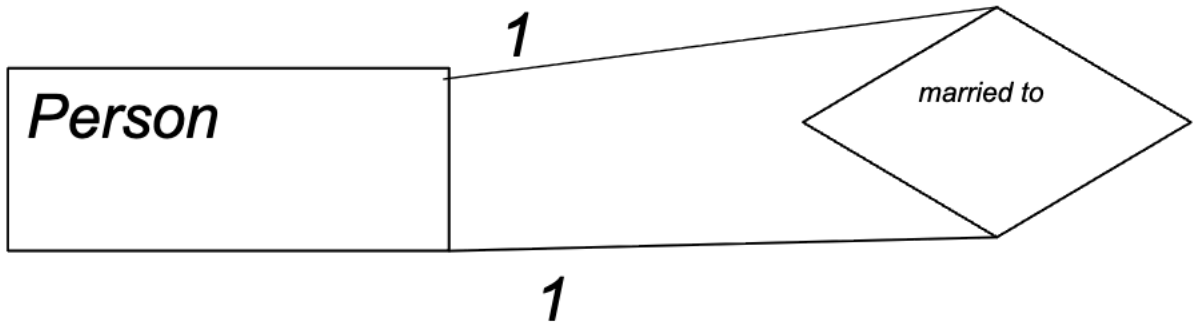
- each entity instance of X may be associated with many entity instances of Y and each entity instance of Y may be associated with with many instances of X

Indicates an instructor may advise many students, and a student may have many advisors.

Remember about the arrow signs!

Recursive Relationship

- An entity can have a relationship with itself!



1:1 Cardinality (Marriage)

Name	ID
Gary	001
Charlie	102
Jade	304
Judy	888
Jian	999

Person (married)

Name	ID
Gary	001
Charlie	102
Jade	304
Judy	888
Jian	999

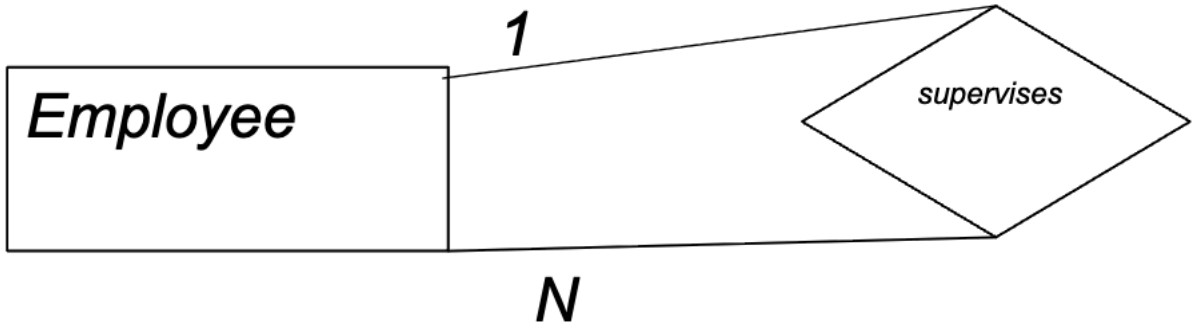
Person (married to)

Coming from same table

Gary is married to Jade
Judy is married to Jian

Recursive Relationship (#2)

- An employee may be the manager of many others, but each employee has only one manager



1:N Cardinality (manager)

Name	ID
Gary	001
Charlie	102
Jade	304
Judy	888
Jian	999

Person (Supervisor)

Name	ID
Gary	001
Charlie	102
Jade	304
Judy	888
Jian	999

Person (Supervised)

Gary can supervise Jade and Judy

Participation Constraints

- Total participation: every entity in the entity set participates in at least one relationship in the relationship set
  - e.g. every course must be taught by a professor
- Partial participation: some entities may not participate in any relationship in the relationship set.
  - e.g. not all professors have to teach courses.

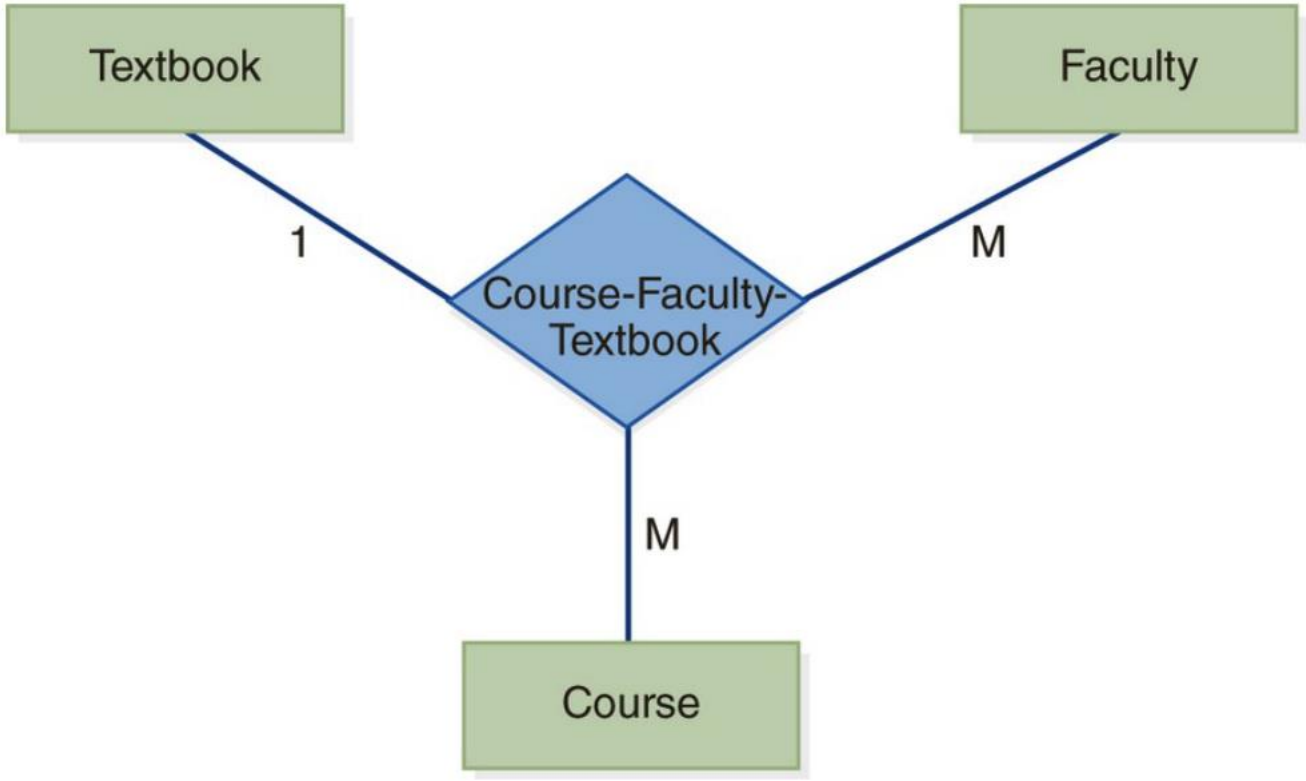
Participation Constraints

- Total participation indicated by a double line (or a bold line)
- Partial participation indicated by a single line

every course must be taught by a professor (total participation)
not all professors have to teach courses. (partial participation)

Ternary Relationship

- A **ternary relationship** is a relationship that involves **three entities at the same time**, where the relationship's meaning depends on all three.



- For a given **(Faculty, Course)** pair, there is **one textbook**.
- A given **(Course, Textbook)** pair can be associated with **many faculty members**.
- A given **(Faculty, Textbook)** pair can be associated with **many courses**.

Book Reference:

- Database system concepts by Silberschatz (6th Edition)
  - Chapter 7